TECHNOLOGY STACK

ComicCraft – AI Comic Story Creator using Gemini Models

Date	15 March 2026	**Team ID**	SWTID-2026-3296
Project Name	ComicCraft – AI Comic Story Creator using Gemini Models	**Team Size**	4

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2 Templates	Used to build the web interface for collecting story prompt, character name, setting, story tone and art style, and to display the generated comic preview and export-success pages. Jinja2 provides dynamic rendering with FastAPI.
2	Backend / Server-Side	Python, FastAPI, Uvicorn	Python provides the application logic; FastAPI manages routing, form/JSON input processing and backend integration; Uvicorn runs the FastAPI application locally as the ASGI server.
3	Database / Data Storage	File-based storage (static/panels and static/exports); no database in the current version	The documented project saves generated panel images under static/panels and exported comic PDFs under static/exports. The current version does not include user account management or a database.
4	Cloud / Hosting / Deployment	Local deployment with Uvicorn	The documented deployment workflow is local: create a Python virtual environment, install requirements, and run the application using Uvicorn. A specific cloud platform is not defined in the project source.
5	Version Control & CI/CD	Git	Git is identified in the project prerequisites for version control and supports maintaining and collaborating on the application source code. No specific CI/CD platform is defined in the project source.
6	Third-Party APIs / Other Tools	Google Gemini APIs; Hugging Face Diffusers / Stable Diffusion; FPDF; Pillow; PyTorch	Gemini Flash generates structured comic outlines; Gemini Pro generates detailed narration and dialogue; Stable Diffusion generates comic-style illustrations. FPDF is used for PDF export, while Pillow, Diffusers and PyTorch support image/model processing.

Team Members

Team Leader – Kamalesh P R Team Member – JAYAKUMARAN V M Team Member – Kannan P Team Member – Naresh C